

# Training-Free RL as Verifiable-Reward Selection on a Single Frozen Speech Model:

Content and Intent Gains, a Scoped Paralinguistic Negative, and a Reward-Spread Lens

Exploring-L4-Intelligence Project

## Abstract

We study *training-free reinforcement learning* on a *single frozen speech model*: no weight updates, no architecture change. Here “training-free RL” means inference-time *verifiable-reward selection*—best-of- $N$ , reranking, and candidate-card retrieval—which we realise concretely with a frozen bi-encoder (omni-embed-nemotron-3b, a SentenceTransformer) that ranks candidate cards by cosine similarity under a verifiable reward. We cast this selection as Gibbs tilting of a frozen base policy,  $q^* \propto q_0 \exp(R/\beta)$ , and read off a single lightweight lens: the gain of selection is exactly  $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$  and is bounded by the *reward spread*  $\text{spread} = \max R - \min R$  over  $\text{supp}(q_0)$ . This is the Donsker–Varadhan / free-energy identity, which we use *qualitatively* rather than as a novel theorem, supported by two strict, sorry-free Lean 4 lemmas: a flat reward yields zero gain (**flat\_no\_gain**), and a non-constant reward yields strictly positive gain (**gain\_pos\_of\_nonconstant**). The lens explains a sharp empirical contrast. Where the verifiable reward has *spread* across candidates—content and intent—selection helps: on MInDS-14 en-US tool/intent, frozen bi-encoder card selection reaches  $\text{Acc}@1 = 0.984$  versus a raw-schema baseline of 0.857, a gain of +0.126 with a 95% paired-bootstrap CI of [0.077, 0.181] ( $n = 182$ , seed 42); SLURP intent corroborates the same mechanism (0.550  $\rightarrow$  0.880). Where the frozen content read-out has *little* paralinguistic spread to exploit, selection does not help: on CREMA-D the same encoder’s frozen embeddings carry near-chance speaker identity (mean acc 0.033,  $3.0\times$  chance, 91-way) and only modest emotion (0.405,  $2.4\times$  chance, 6-way), and training-free pool selection yields no significant emotion gain (across-seed 95% CI  $[-0.043, +0.116]$ , spanning zero). We treat this precise, committed negative as a first-class contribution. All positive and negative results are frozen bi-encoder selection with committed reproduction artifacts; the broader agent-level program is deferred as future work.

## 1 Introduction

We ask how much can be extracted from a *single frozen speech model* without touching a weight or changing its structure, using reward-guided optimization applied purely at inference time. A modern omni or speech-capable multimodal LLM is, by release, an extraordinary repository of latent capability: from large-scale audio, parallel speech-text corpora, and instruction mixtures, models such as Qwen2-Audio, Qwen2.5-Omni, Qwen3-Omni, and SALMONN [qwe, 2024, 2025a,b, sal, 2023] acquire transcription, translation, speaker identity, paralinguistic affect, and dialogue competence simultaneously. Our question is deliberately conservative: *how far can one go with the model held fixed?* We treat “training-free RL” concretely as inference-time *verifiable-reward selection*—best-of- $N$  selection, reranking, and candidate-card retrieval—that changes no parameters and no architecture [bei, 2024, mud, 2023, zuo, 2025]. In all of our experiments this selection is realised by a *frozen bi-encoder* (omni-embed-nemotron-3b, a SentenceTransformer) that ranks candidate cards by cosine similarity under a verifiable reward. We never train, fine-tune, or run a local generative operator; every result below is frozen-encoder selection.

**The lever is the reward spread.** The premise that organizes this paper is that the gain available to inference-time selection is a property of *which reward, over which candidates* one selects. Some rewards are nearly *flat* over the candidate set the frozen model actually places mass on: selection then moves through a narrow envelope of near-identical rewards and cannot manufacture headroom the candidate distribution does not contain. For speech in particular, a content encoder tends to “read” a transcribable surface rather

than “listen” to the acoustic substrate, so its embeddings expose wide spread for content and intent but little spread for paralinguistics [min, 2022, vox, 2026, hsu, 2023]. The central object of the paper is this reward *spread*, and the empirical fact that it is large for content/intent and small for paralinguistics on one and the same frozen encoder.

**Selection as Gibbs tilting.** To reason about this quantitatively we adopt the standard variational view of inference-time alignment. Optimizing the KL-regularized objective

$$F(q) = \mathbb{E}_{z \sim q}[R(z)] - \beta \text{KL}(q \parallel q_0)$$

over selection distributions  $q$  on the candidate space  $\mathcal{Z}$  has the closed-form Gibbs optimum  $q^*(z) \propto q_0(z) \exp(R(z)/\beta)$ , with partition function  $Z$  [kor, 2022, bei, 2024]. Every practical training-free selector—best-of- $N$  and its soft refinements [ver, 2025, yan, 2024, gui, 2024], self-consistency [wan, 2022], MBR/reranking [ich, 2025, mbr, 2025]—is an estimator of this tilt over some realization of  $\mathcal{Z}$ . The attainable benefit of any such estimator is bounded by how far  $q^*$  can depart from  $q_0$ , which is governed by the spread of  $R$  over  $\text{supp}(q_0)$ .

**A reward-spread lens.** Our lens is a single, lightweight identity, stated at its true strength rather than as a discovery. Granted the Gibbs framing, the gain is exactly  $\mathcal{G} = F(q^*) - F(q_0) = \beta \text{KL}(q_0 \parallel q^*)$ , and a Hoeffding argument caps it by the reward spread,  $\mathcal{G} \leq \text{spread}^2/(8\beta)$ ; to leading order the discriminator sharpens to  $\mathcal{G} \approx \text{Var}_{q_0}(R)/(2\beta)$ . This is the Donsker–Varadhan / free-energy identity, and we are explicit that we use it *qualitatively*, to explain the content-versus-paralinguistic contrast, not as a novel theorem. Its two load-bearing corners are machine-checked sorry-free in Lean 4: a flat reward yields zero gain (`flat_no_gain`), and a non-constant reward yields strictly positive gain (`gain_pos_of_nonconstant`). The reading is direct: high reward spread  $\Rightarrow$  selection can help; low spread  $\Rightarrow$  it cannot.

**Relation to prior training-free work.** Training-free selection—best-of- $N$ , reranking, MBR—is established, most visibly for ASR and open-ended generation [bei, 2024, mbr, 2025]. This paper is deliberately narrow and different: we apply a *frozen omni-embed encoder* to spoken-language-understanding and intent *selection*, we report a precise *paralinguistic negative* on the same encoder, and we read both through the reward-spread lens. We do *not* claim novelty in ASR best-of- $N$ ; our contribution is the SLU/intent selection result, the committed negative, and the lens that ties them together.

**Contributions.** We make exactly three.

**(C1) A real, local, reproducible content/intent selection gain.** Where the verifiable reward has spread across candidates, frozen bi-encoder card selection helps. On MInDS-14 en-US tool/intent it reaches  $\text{Acc}@1 = 0.984$ ; against a raw-schema baseline of 0.857 this is a gain of +0.126 with a 95% paired-bootstrap CI of [0.077, 0.181] ( $n = 182$ , seed 42; committed artifact `_repro/minds14_toolintent_paired.json`, with the reproduction command committed). SLURP intent corroborates the same frozen-bi-encoder card-selection mechanism (0.550  $\rightarrow$  0.880, +0.330). Training-free selection helps precisely because the reward has spread here.

**(C2) A precise, committed paralinguistic negative.** The frozen omni content read-out has low paralinguistic spread to exploit. On CREMA-D (dev-selected layer, 3 seeds; committed `_repro/paralinguistic_negative_probe`) the frozen embeddings carry near-chance speaker identity (mean acc 0.033,  $3.0\times$  chance, 91-way) and only modest emotion (0.405,  $2.4\times$  chance, 6-way); and training-free pool selection yields *no* significant emotion gain (across-seed 95% CI  $[-0.043, +0.116]$ , spanning zero; committed `_repro/emotion_pool_paired_v2.json`). We treat this as a first-class contribution: a precise measurement plus the spread explanation for why selection over this encoder cannot lift paralinguistics.

**(C3) A lightweight reward-spread lens.** We formalize the gain identity  $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$  and its spread ceiling, use it qualitatively as the Donsker–Varadhan / free-energy identity, and support it with two strict, sorry-free Lean 4 lemmas (`flat_no_gain`, `gain_pos_of_nonconstant`). The lens explains the C1–C2 contrast: content/intent has high reward spread and selection gains; paralinguistics has low spread and no gain.

**Scope and deferred work.** Everything we report is frozen bi-encoder verifiable selection over candidate cards with committed reproduction artifacts; we do not claim agent-level, multi-session, or generative-operator results. The broader agent-level program is genuinely future work: by the spread lens, context isolation over a fixed reward adds nothing, so making such a program meaningful would require (a) a cross-session same-speaker corpus, absent from our frozen data, (b) a locally-driven generative operator we do not build here, and (c) a genuinely new non-separable irreducibility result rather than the separable identity we use. We under-claim deliberately.

**Organization.** Section 1 has framed the reward-spread lens and the three contributions. The remainder develops the Gibbs-tilt objective and the spread identity, with the two sorry-free Lean lemmas; presents the frozen bi-encoder selection setup and the MInDS-14/SLURP content-intent results (C1); reports the CREMA-D paralinguistic probe and the null emotion-selection result (C2); and closes with related work, limitations, and the deferred agent-level program.

## 2 Preliminaries and Problem Formulation

We study *training-free* reinforcement learning (RL): a regime in which a single frozen multimodal speech/omni model is steered at inference time by a reward signal, with no gradient ever propagated into its parameters and no architectural surgery. In this paper the regime is realized concretely as *verifiable-reward selection*—best-of- $N$ , reranking, and candidate-card retrieval—over a fixed candidate pool, using a frozen bi-encoder to score candidates. This section fixes notation, states the invariant that distinguishes training-free RL from conventional alignment, specifies the single inference-time operator we actually use, introduces the reward-spread quantity that governs when selection helps, and specifies the reward class we admit. The objects defined here are reused verbatim in the Theory section.

### 2.1 Training-free RL as constrained inference-time reward maximization

Let  $g_\theta$  denote a pretrained speech/omni model with *frozen* parameters  $\theta$ , and let  $x$  be an input (an utterance, a prompt, or a paired context). The inference-time degrees of freedom—the only quantities we are permitted to vary—are collected into an *action*  $z \in \mathcal{Z}$ , where  $\mathcal{Z}$  is the selection action space made precise in Section 2.2. Executing an action produces an output that is scored by a bounded scalar *reward*  $R: \mathcal{Z} \rightarrow [0, 1]$  (we write  $R(z)$  for the reward of the output realized by  $z$  when  $x$  is fixed). A pretrained *base* (or prior) distribution  $q_0 \in \Delta(\mathcal{Z})$  over actions is induced by the frozen model and the default inference procedure. Training-free RL searches over distributions  $q \in \Delta(\mathcal{Z})$  for one that raises expected reward while remaining close to the prior.

**Definition 1** (Training-free RL objective). Fix  $x$ , a base distribution  $q_0$ , a reward  $R$ , and a temperature  $\beta > 0$ . The KL-regularized inference-time objective is

$$F(q) = \mathbb{E}_{z \sim q}[R(z)] - \beta \text{KL}(q \parallel q_0), \quad (1)$$

and the training-free RL problem is  $q^* = \arg \max_{q \in \Delta(\mathcal{Z})} F(q)$ .

The objective (1) is strictly concave in  $q$  over the simplex, and its maximizer is the classical Gibbs/exponential tilt of the prior [kor, 2022].

**Proposition 1** (Gibbs optimum). *The unique maximizer of (1) is*

$$q^*(z) = \frac{1}{Z} q_0(z) \exp\left(\frac{1}{\beta} R(z)\right), \quad Z = \mathbb{E}_{z \sim q_0}\left[\exp\left(\frac{1}{\beta} R(z)\right)\right], \quad (2)$$

with optimal value  $F(q^*) = \beta \log Z$ .

*Proof sketch.* Adding a Lagrange multiplier for the normalization constraint and differentiating the concave functional (1) with respect to  $q(z)$  yields  $R(z) - \beta(\log q(z) - \log q_0(z)) - \beta = \text{const}$ , i.e.  $\log q(z) = \log q_0(z) + R(z)/\beta + \text{const}$ ; exponentiating and normalizing gives (2). Substituting back collapses the objective to  $\beta \log Z$ .  $\square$

Equation (2) is the formal anchor of the paper: every concrete inference-time procedure we analyze—best-of- $N$ , soft best-of- $N$ , reranking, and candidate-card selection—is read as an *estimator of, or a sampler from, the tilted target*  $q^*$ , and is therefore comparable through the lens of how well it approximates (2) at a given query budget. The temperature  $\beta$  interpolates between the prior ( $\beta \rightarrow \infty$ ,  $q^* \rightarrow q_0$ ) and greedy reward maximization ( $\beta \rightarrow 0^+$ ,  $q^* \rightarrow \text{the } \arg \max_z R \text{ point mass}$ ), recovering best-of- $N$  selection in the low-temperature limit [bei, 2024, ver, 2025, kha, 2025].

**Definition 2** (No-weight-update invariant). A procedure is *training-free* if, for every input and every query budget, the model parameters  $\theta$  and the model’s computational graph are held fixed; all adaptation is realized through the choice of  $q$  over the inference-time action space  $\mathcal{Z}$ . Formally, the map  $\theta \mapsto \theta$  is the identity throughout, and only  $q$  is optimized.

Definition 2 is the operational contract of this work. It excludes parameter-efficient fine-tuning and any backpropagation through  $g_\theta$ ; it admits only *selection*, *reweighting*, and *reranking* over a fixed candidate pool. Because  $q^*$  depends on  $\theta$  only through  $q_0$  and on the task only through  $R$ , the frozen model’s pretrained competence is *activated*, never overwritten—the central premise the paper develops empirically and theoretically.

## 2.2 The selection operator: a frozen bi-encoder ranking candidate cards

Every reproducible result in this paper is produced by a single inference-time operator: a frozen bi-encoder that ranks a finite pool of candidate *cards* by cosine similarity under a verifiable reward. We make this precise; no other operator is instantiated in our experiments.

For a query  $x$  we are given a finite *candidate pool*  $C(x) = \{y_1, \dots, y_N\}$  (e.g., tool/intent schema cards, or transcript hypotheses). The action space is the probability simplex over this pool,  $\mathcal{Z} = \Delta(C(x))$ ; an action  $z$  is a distribution over candidates, and a *selection* action collapses to the vertex that returns a single chosen card.

**Definition 3** (Frozen bi-encoder selection operator). Let  $E$  be a frozen bi-encoder (in our experiments `omni-embed-nemotron-3b`, a SentenceTransformer) that maps the query and each candidate card to  $\ell_2$ -normalized vectors  $E(x)$  and  $E(y_i)$ . The operator scores each candidate by cosine similarity  $s_i = \langle E(x), E(y_i) \rangle$  and induces the base distribution  $q_0(y_i) \propto \exp(s_i/\beta)$  over the pool. Training-free RL reweights  $q_0$  toward the verifiable reward via the Gibbs tilt (2), and the realized *selection* returns  $\arg \max_i R(y_i)$  over the ranked pool (best-of- $N$ ), or the top-ranked card when the reward is used to construct the pool rather than to pick from it.

Two properties of Definition 3 matter for the analysis. First, the operator is a *single-sample deterministic readout* over a geometric object (points on the unit sphere): it adds no output entropy of its own and can only reweight candidates the pool already contains. Second, its usefulness is entirely governed by how much the reward *varies across the pool*: if the reward is nearly constant over  $C(x)$ , selection cannot improve on the prior, whereas a reward that separates good from bad candidates lets selection concentrate mass on the reward maximizer. We make this dependence explicit next.

*Remark 1* (Generative operators are out of scope). A co-equal generative operator—an autoregressive thinker–talker model that *produces* rather than *selects* candidates, with best-of- $N$  or reward-guided decoding tilting its sampler toward (2)—is a natural extension [qwe, 2025b, mud, 2023]. We do not instantiate one locally, and we make no claim about it; it is discussed only as future work. All results below concern the frozen bi-encoder selection operator of Definition 3.

## 2.3 Reward spread

Selection helps exactly to the extent that the reward is *not flat* over the candidate pool. We quantify this with the reward spread, the object that the qualitative lens in the Theory section (OSA-1) ties to the optimization gain.

**Definition 4** (Reward spread and optimization gain). Fix  $x$ ,  $q_0$ ,  $R$ , and  $\beta$ . The *reward spread* spread is the range of  $R$  over the support of the prior,  $\text{spread} = \max_{z \in \text{supp}(q_0)} R(z) - \min_{z \in \text{supp}(q_0)} R(z)$  (to leading

order the  $q_0$ -weighted variance  $\text{Var}_{q_0}(R)$  plays the same role), and the *optimization gain* of training-free RL over the prior is

$$\mathcal{G} = F(q^*) - F(q_0) = \beta \log Z - \mathbb{E}_{z \sim q_0}[R(z)] \geq 0, \quad (3)$$

with equality if and only if  $R$  is  $q_0$ -almost-surely constant.

Equation  $(\star)$  is the free-energy expression for the value of the tilt:  $\mathcal{G}$  is nonnegative, vanishes exactly when the reward carries no spread, and grows as the reward separates candidates. The Theory section develops this as a qualitative lens (the Donsker–Varadhan / free-energy identity, used to explain—not to newly prove—why gains appear) and pins its two endpoints with strict, machine-checked lemmas: a flat reward yields zero gain (`flat_no_gain`), and a non-constant reward yields strictly positive gain (`gain_pos_of_nonconstant`). Empirically this is the reading of our contrast: content/intent tasks expose high reward spread across candidate cards and therefore admit large selection gains, whereas the frozen model’s paralinguistic read-out exposes little spread and selection yields no significant gain.

## 2.4 Verifiable versus model-judged rewards

The validity of every guarantee in Sections 2.1 and 2.3 rests on  $R$  being trustworthy. We admit only *verifiable* rewards.

**Definition 5** (Verifiable reward). A reward  $R$  is *verifiable* if it is a deterministic, label-derived function of the output and a ground-truth annotation  $y^*$ , computable without invoking a learned scorer. Canonical instances are (i) negative word error rate  $1 - \text{WER}(y, y^*)$  for ASR/ST, (ii) exact-match/accuracy for classification and intent, and (iii) retrieval hit@ $k$  / probe accuracy against a fixed label. A reward is *model-judged* if it is the output of a separate generative or preference model evaluating  $y$ .

**Assumption 1** (Verifiability). Throughout,  $R$  is verifiable in the sense of Definition 5: bounded in  $[0, 1]$ , deterministic given  $y^*$ , and independent of  $g_\theta$ .

We adopt Assumption 1 for two reasons. First, model-judged rewards are susceptible to *reward hacking*—the optimizer exploits gaps between the proxy and the true objective, so that increasing  $\mathbb{E}_{q^*}[R]$  no longer increases task quality [ska, 2022, gao, 2022]. Because the tilt (2) sharpens precisely along the direction of  $R$ , any reward misspecification is amplified rather than averaged out as  $\beta \rightarrow 0$ . Second, learned judges carry systematic artifacts—position bias and self-preference—that contaminate the candidate ranking [pos, 2024, sel, 2024]; verifiable rewards are invariant to these. Restricting to label-derived  $R$  is therefore not a convenience but a precondition for the no-weight-update invariant of Definition 2 to translate into genuine, contamination-free activation of pretrained competence [tul, 2024]. With Definitions 1 to 5 and Assumption 1 in place, the notation required by the Theory section is complete.

## 3 Related Work

Our contribution sits at the confluence of three literatures: the theory of inference-time reward optimization as exponential tilting, the empirical study of in-context learning (ICL) in audio language models, and the geometric characterization of what frozen speech encoders represent. We review each in turn, emphasizing not a catalogue of methods but a single structural fact that our experiments make concrete: training-free RL, realized here as *verifiable-reward selection* over a fixed candidate set  $\mathcal{Z}$  drawn from one frozen speech model, is bounded by the *spread* of the reward across those candidates. Where the reward spreads—content and intent selection over candidate cards—selection buys real gains; where it is nearly flat—paralinguistic factors the frozen read-out has largely pooled away—selection has nothing to exploit. This spread lens organizes both our positive result (frozen bi-encoder card selection for spoken-language understanding) and our committed negative (little training-free paralinguistic gain over the same encoder).

### 3.1 Inference-Time / Training-Free RL and the Tilting View

The unifying abstraction behind training-free RL is that a reward-guided search distribution  $q$  is, in its optimal form, an exponential tilt of the base policy: the KL-regularized objective  $F(q) = \mathbb{E}_q[R] - \beta \text{KL}(q \parallel q_0)$  is



maximized by the Gibbs measure  $q^*(z) \propto q_0(z) \exp(R(z)/\beta)$ , with partition function  $Z$  [kor, 2022]. This identity recasts a spectrum of seemingly disparate inference-time procedures as approximations to the same tilt, and it is the abstraction under which our own method—selecting the best candidate card under a verifiable cosine reward—sits. Best-of- $N$  (BoN) sampling is the most studied instance: bei [2024] establish that the KL divergence between the BoN distribution and  $q_0$  grows as  $\log N - (N-1)/N$ , giving a closed-form exchange rate between compute and the distributional shift purchased. ver [2025] sharpen this with a soft, temperature-controlled BoN whose suboptimality decays as  $O(1/N)$ , while ami [2025] recast BoN through a smoothing lens, bounding its regret against the Gibbs optimum and exposing when additional samples cease to help. The crucial—and for our purposes load-bearing—observation across this line is that *every* tilt reweights existing candidates: the support of  $q$  is contained in the support of  $q_0$ . No amount of  $N$  creates mass where  $q_0$  assigns none, and, more sharply, no reweighting helps when the reward is constant across the candidate set. Minimum Bayes risk (MBR) decoding [ich, 2025], including its specialization to ASR [mbr, 2025], and controlled decoding [mud, 2023] inherit the same ceiling: they are consensus or value-guided rerankings, powerful when a correct hypothesis is already present with non-negligible probability and inert otherwise.

A more recent strand attempts to extract reward signal without any labels at all. TTRL [zuo, 2025] and TPO [li2, 2025] use self-consistency or majority vote as a surrogate reward at test time, and JitRL [jit, 2026] performs gradient-free credit assignment over a frozen policy. These methods are attractive precisely because they touch no weights, but they sharpen rather than dissolve the support constraint: a self-generated reward can only amplify modes the model already prefers, which is why majority-vote rewards can entrench confident errors. The over-optimization literature quantifies the resulting hazard. gao [2022] show that reward and true objective diverge as KL grows, following a predictable scaling law; kha [2025] characterize the optimal effective  $N^*$  (the Best-of-Poisson regime) beyond which tilting harms. In our notation these dynamics are governed by the reward spread and the realized identity gain  $\mathcal{G}$ : the achievable gain is  $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$ , which vanishes exactly when the reward is flat and grows with its spread. This is the qualitative content of the Donsker–Varadhan / free-energy identity we use throughout—not as a novel theorem, but as a lens—and it is the same quantity a verifiable-reward selector is implicitly maximizing over its candidate cards.

These dynamics intersect the ongoing RLVR boundary debate, which directly concerns whether inference-time or verifiable-reward optimization can expand a model’s capabilities or merely re-allocate them. rlv [2025] argue that RLVR predominantly sharpens the base distribution rather than adding skills, a position echoed by the finding that even spurious or random rewards can improve benchmark scores by exploiting format and prior structure [spu, 2025]. The countervailing evidence that RLVR can induce genuinely new skills [new, 2026] does not contradict the tilting view so much as locate its exception: new behavior emerges only when the candidate set  $\mathcal{Z}$  carries enough reward spread—i.e., when a better hypothesis had non-zero base mass that a default single-shot read-out rarely surfaced. This reweighting-versus-expansion axis is exactly the one our spread lens sharpens, and it predicts, correctly for our data, where frozen selection helps (high-spread content/intent) and where it does not (near-flat paralinguistics).

### 3.2 In-Context Learning in Audio LLMs

A natural alternative to reward-guided selection is to move the base distribution with in-context demonstrations, and the text-domain ICL literature offers a sobering prior. min [2022] demonstrate that demonstrations are largely label-insensitive: corrupting ground-truth labels barely degrades performance, implying that demonstrations specify the task and output format rather than teaching the input–output mapping. ali [2026] reach a congruent conclusion for instruction-following demonstrations, which fix format but not accuracy. ICL is moreover scale-gated—its more sophisticated, override-the-prior behaviors emerge only in larger models [wei, 2023]—a property that carries into audio, where few-shot competence appears as an emergent phenomenon in larger audio LLMs [mim, 2025]. The function- and task-vector view [tod, 2023, tas, 2025] explains the mechanism: demonstrations compress to a latent task direction that conditions decoding without altering the underlying perceptual representation. In-context steering can thus re-route *how* content is verbalized, but it does not re-open perceptual channels the encoder has already discarded.

This perceptual ceiling is starkly visible in audio. A growing body of evidence shows that audio LLMs frequently “read rather than listen”: vox [2026] document that models answer paralinguistic questions from

textual priors and transcript content rather than the acoustic signal, and [lis \[2025\]](#) find that performance often depends on whether the model transcribes versus genuinely attends to non-lexical cues. In-context demonstrations do little to repair this when the requisite acoustic information has already been pooled away upstream—consistent with the broader finding that few-shot prompting fixes format, not the missing percept. SER is a partial and instructive exception: [ser \[2025\]](#) report that few-shot demonstrations *do* improve speech emotion recognition, plausibly because emotion is one of the paralinguistic factors an omni model retains with enough base probability to be surfaced. The pattern—ICL helping for retained factors, failing for discarded ones—mirrors the spread constraint of Section 3.1, and it foreshadows our committed negative: over a frozen content-oriented encoder, the paralinguistic reward has little spread for any training-free procedure, selection or ICL, to exploit.

The contrast with embedding models is relevant because our method uses precisely such a model—a frozen bi-encoder—as the selector. Text embedders do not acquire ICL for free: [bge \[2024\]](#) show that exploiting in-context examples in an embedder requires dedicated training, and instruction-conditioned embedders such as INSTRUCTOR [\[su2, 2022\]](#) must be explicitly trained to follow task instructions at encode time. An off-the-shelf embedding encoder therefore exposes no prompt-conditioned degree of freedom: its available action space is the choice of query, pooling, and layer, plus which candidate card to rank first—not a re-conditioning of the forward pass. Our selector operates entirely within this space, which is why the reward spread available in the encoder’s fixed geometry is the binding constraint on what training-free selection can deliver.

### 3.3 Embedding Geometry and Probing

Whether a frozen encoder *can* be tilted toward a task ultimately depends on whether the relevant information survives in its representations—a question the probing and embedding-geometry literatures answer with unusual clarity, and one that underpins our paralinguistic negative. Contrastive and alignment objectives are now understood to actively discard nuisance variation: [xia \[2020\]](#) show that contrastive learning suppresses factors not aligned with the training augmentations, and [wan \[2020\]](#) formalize the alignment–uniformity trade-off that drives semantically similar inputs together while spreading the hypersphere uniformly. For a content-oriented encoder, speaker identity, emotion, and channel are precisely the nuisances such objectives are designed to remove. This is the geometric root of why a frozen selector, however cleverly searched, cannot recover a factor its training objective erased: the base distribution  $q_0$  over the candidate set simply carries no reward spread along the discarded direction.

Pooling compounds the loss. Mean- or last-token pooling imposes an information bottleneck that NV-Embed’s learned latent-attention pooling was introduced to mitigate [\[nve, 2024\]](#), and the speaker-verification community has long known that attentive statistics pooling [\[att, 2018\]](#) and ECAPA-TDNN’s multi-scale aggregation [\[eca, 2020\]](#) are necessary precisely because naive temporal averaging destroys speaker-discriminative structure. The implication is that the *same* encoder can be near-blind or highly informative for a factor depending solely on the read-out. Layer-wise probing makes this concrete: [pas \[2021\]](#) chart an autoencoder-like trajectory across wav2vec 2.0 layers in which phonetic and word-level information peaks at intermediate depths, and [spk \[2025\]](#) show speaker attributes are distributed non-uniformly across layers. Capability is thus not a scalar property of a model but a function of (layer, pooling, query). Our committed probe of the frozen omni content read-out is consistent with this picture from the pessimistic side: on CREMA-D it recovers speaker identity only near chance and emotion only modestly, so a training-free selector operating on that read-out inherits a near-flat paralinguistic reward and, empirically, no significant emotion gain.

Evidence that paralinguistic content *persists* elsewhere in a model, even when unused, is nonetheless substantial. [fro \[2024\]](#) demonstrate that a frozen LLM, given speech features, perceives paralinguistic attributes it never verbalizes; [emo \[2026\]](#) localize emotion-selective units inside such models. We read these results carefully: they show the information can survive in *some* read-out, not that it is available in the particular content-oriented encoder our selector uses, and our negative quantifies exactly that gap for one committed configuration. The SUPERB and Dynamic-SUPERB taxonomies [\[sup, 2021, dyn, 2023\]](#) supply the benchmark scaffolding for evaluating capability across content, speaker, semantic, and paralinguistic axes, and we adopt their factorization to structure our measurements. What this literature lacks—and what our

spread lens supplies—is a principled account of *which* latent capabilities are reachable by reweighting a fixed candidate set and which demand a richer read-out or a different action space altogether.

### 3.4 Cross-Session Agent Memory and Skills (Future-Work Context)

Our contributions are confined to a single frozen speech model and inference-time verifiable-reward *selection*; we make no agent-level claim. For completeness we note the literature that would motivate a *deferred*, out-of-scope program on cross-session speech agents, and we are explicit that none of it is related to a result we report here. A line of work on training-free agent memory accumulates experience without touching weights — from raw interaction streams and paged context [gen, 2023, mem, 2023] to first-class curation operators [mem, 2025a], with reinforcement learning used to assign credit over memory-management actions [mem, 2025b]; a parallel line accumulates reusable, reward-verified skills atop frozen models [voy, 2023, sok, 2026]. Evaluation of such systems remains almost entirely text-centric and single-episode: long-horizon and conversational probes exist [lon, 2024, loc, 2024], audits show many ostensibly memory-dependent tasks need no genuine cross-session recall [ana, 2026], and spoken benchmarks evaluate a fresh agent per interaction [uro, 2025, voi, 2024] with no isolated cross-session, same-speaker suite [amb, 2026].

We flag this only to scope our reward-spread lens honestly. Extending selection to an agent that persists across sessions would require three ingredients absent from our setting: (a) a cross-session, same-speaker corpus (none exists in our frozen data); (b) a locally-driven generative operator (we run only a frozen bi-encoder over candidate cards, never a local generative model); and (c) a genuinely new, non-separable irreducibility result. Absent (c), the spread lens implies context isolation adds nothing on a fixed reward: a memory that merely re-partitions the candidate pool cannot raise the identity gain beyond what the reward spread already permits. We therefore leave the agent-level program to future work and claim none of it here.

## 4 A Reward-Spread Lens

We keep one lightweight analytic tool, used *qualitatively* to organize the empirical contrast between content/intent and paralinguistics — not as a novel theorem, and not as a “machine-checked quantitative ceiling.” The claim is simple: inference-time verifiable-reward selection over a frozen model can only realize gain to the extent that the reward actually *spreads* across the candidates it ranks. Where the reward has spread (content, tool/intent), selection helps; where it is nearly flat (speaker, emotion on our frozen omni read-out), it cannot.

Fix a finite candidate set  $\mathcal{Z}$  (the candidate cards a frozen bi-encoder ranks), a strictly positive base measure  $q_0 : \mathcal{Z} \rightarrow \mathbb{R}_{>0}$  with  $\sum_{w \in \mathcal{Z}} q_0(w) = 1$ , a bounded verifiable reward  $R : \mathcal{Z} \rightarrow \mathbb{R}$ , and an inverse-temperature  $\beta > 0$ . Write  $\mathbb{E}_q[\cdot]$  for expectation under  $q$ , adopt  $0 \log 0 = 0$ , and define the tilted (Gibbs) target and its normalizer

$$Z := \mathbb{E}_{q_0}[\exp(R/\beta)] > 0, \quad q^*(z) := \frac{q_0(z) \exp(R(z)/\beta)}{Z},$$

together with the KL-regularized objective  $F(q) := \mathbb{E}_q[R] - \beta \text{KL}(q \parallel q_0)$ . This is the standard inference-time specialization of KL-regularized reward maximization, whose maximizer is the exponential tilt [kor, 2022, mud, 2023].

**Master identity: gain is a KL divergence.** A one-line algebraic identity (substitute  $R(z) = \beta(\log \frac{q^*(z)}{q_0(z)} + \log Z)$  into  $F(\cdot)$ ) gives, for every distribution  $q$  on  $\mathcal{Z}$ ,

$$F(q) = \beta \log Z - \beta \text{KL}(q \parallel q^*).$$

Reading it at the model’s own distribution  $q = q_0$  and defining the optimization *gain*  $\mathcal{G} := F(q^*) - F(q_0)$  yields the object we actually care about:

$$\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*) = \beta \log \mathbb{E}_{q_0}[\exp(R/\beta)] - \mathbb{E}_{q_0}[R]. \quad (\star)$$

The gain is exactly the divergence between what the model already does,  $q_0$ , and what the reward would have it do,  $q^*$ . The right-hand form of  $(\star)$  is the free-energy / cumulant expression  $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R]$ ;



by the Donsker–Varadhan variational identity it is nonnegative, and it is governed by how much  $R$  varies under  $q_0$ . We use this *qualitatively*: a second-order expansion gives  $\mathcal{G} \approx \text{Var}_{q_0}(R)/(2\beta)$  for a concentrated  $q_0$ , so the gain scales with the *spread* spread of the reward across candidates. This is a well-known identity, invoked here as an organizing lens, not billed as a new result or a tight ceiling.

**Two endpoints (both sorry-free in Lean).** The lens has two exact endpoints, which we have formalized in Lean 4 without `sorry`.

(i) *Flat reward  $\Rightarrow$  zero gain* (`flat_no_gain`). If  $R \equiv c$  is constant on  $\mathcal{Z}$ , then  $Z = e^{c/\beta}$ , so  $q^* = q_0$  and, by  $(\star)$ ,  $\mathcal{G} = 0$ . A reward that cannot distinguish the candidates leaves the model unchanged: there is nothing for selection to exploit.

(ii) *Non-constant reward  $\Rightarrow$  strictly positive gain* (`gain_pos_of_nonconstant`). If  $R$  is non-constant on  $\mathcal{Z}$  (recall  $q_0 > 0$  everywhere), then  $q_0 \neq q^*$ : were they equal,  $e^{R(z)/\beta} = Z$  for all  $z$  would force  $R$  constant. By the strict Gibbs inequality  $\text{KL}(q_0 \parallel q^*) > 0$ , so  $(\star)$  gives  $\mathcal{G} > 0$ . Any genuine spread in the verifiable reward yields strictly positive headroom for selection to realize.

These two lemmas are the strict converse of one another:  $R \text{ constant} \iff q^* = q_0 \iff \mathcal{G} = 0$ .

**Why the lens explains our results.** The lens reads directly onto the empirical contrast of this paper. For *content and intent*, the verifiable reward separates candidate cards sharply: on MInDS-14 en-US tool/intent, frozen bi-encoder candidate-card selection reaches  $\text{Acc}@1 = 0.984$  versus the raw-schema baseline 0.857 ( $\Delta = +0.126$ ; Table 1), and SLURP intent corroborates ( $0.550 \rightarrow 0.880$ ). Here  $R$  has high spread across candidates, so by  $(\star)$  and endpoint (ii) selection realizes a real gain — the inference-time analogue of why best-of- $N$  and reranking pay off in proportion to how much a verifiable reward separates candidates [bei, 2024, gao, 2022]. For *paralinguistics*, the picture is the opposite. The frozen omni content read-out carries near-chance speaker information and only modest emotion (Table 2), and training-free pool-selection produces no significant emotion gain (across-seed 95% CI  $[-0.043, +0.116]$ ; committed artifact `_repro/emotion_pool_paired_v2.json`). On these factors  $R$  is nearly flat on  $\text{supp}(q_0)$ , so by endpoint (i) there is little spread to exploit and selection cannot help — exactly what we observe. The reward-spread lens thus does one job, and only that job: it explains why identical training-free selection machinery succeeds on content/intent and fails on paralinguistics, without appeal to model class, decomposition, or agentic structure.

## 5 Empirical Results

We report two results on a *single frozen* speech model, one positive and one negative, and read both through a single quantitative lens—the *spread* of the verifiable reward across candidates. All “training-free RL” here is inference-time *verifiable-reward selection*: a frozen bi-encoder (`omni-embed-nemotron-3b`, a SentenceTransformer) ranks candidate cards by cosine similarity under a verifiable reward, realising best-of- $N$  / reranking / candidate-card retrieval with *no weight updates and no structural change*. The positive result is a content/intent *selection* gain where the reward has large spread across candidates (Section 5.2); the negative is a paralinguistic read-out and a null emotion-selection gain, where the reward has little spread to exploit (Section 5.3). The two are explained by the same lens, which we state first (Section 5.1).

### 5.1 The Reward-Spread Lens

The optimization we perform is the KL-regularised tilt of the frozen base distribution  $q_0$  toward a verifiable reward  $R$ : maximising  $F(\cdot \parallel q) = \mathbb{E}_q[R] - \beta \text{KL}(\cdot \parallel q_0)$  has optimum  $q^* \propto q_0 \exp(R/\beta)$ , the same object that underwrites best-of- $N$  and reward-guided decoding [kor, 2022, mud, 2023, bei, 2024], here applied to selection among candidate cards. The gain of this tilt admits the identity

$$\mathcal{G} = \beta \text{KL}(\cdot \parallel q_0 \parallel q^*),$$

which we use *qualitatively*: it is the Donsker–Varadhan / free-energy identity for the log-partition function  $\log Z = \log \mathbb{E}_{q_0}[\exp(R/\beta)]$ , not a novel theorem of ours. Its content for us is a comparison, not a constant: the

Table 1: Frozen bi-encoder verifiable-reward *selection* on intent (single frozen **omni-embed-nemotron-3b**, cosine ranking of candidate cards; seed 42). MInDS-14 en-US tool/intent,  $n=182$ : the policy card improves over the raw-schema baseline by +0.126, 95% paired-bootstrap CI [0.077, 0.181] (committed artifact `_repro/minds14_toolintent_paired.json`). SLURP intent corroborates the same card-selection mechanism. These are selection gains, not generative decoding.

| Task ( $n$ )   | Candidate-card representation | Acc@1 | $\Delta$ vs. baseline | 95% CI         |
|----------------|-------------------------------|-------|-----------------------|----------------|
| MInDS-14 (182) | naive card                    | 0.720 | —                     | —              |
|                | raw-schema (baseline)         | 0.857 | —                     | —              |
|                | policy (boundary-aware)       | 0.984 | +0.126                | [0.077, 0.181] |
| SLURP intent   | basic-label (baseline)        | 0.550 | —                     | —              |
|                | boundary card (policy)        | 0.880 | +0.330                | —              |

gain is controlled by the reward *spread*  $\text{spread} = \max R - \min R$  over  $\text{supp}(q_0)$ . A flat reward—one that does not discriminate among candidates—yields *zero* gain; a reward with genuine spread across candidates yields *strictly positive* gain, and does so *on a single frozen model*, with no decomposition or additional structure required. Two strict, **sorry-free** Lean lemmas make exactly these two endpoints precise: **flat\_no\_gain** (a constant reward gives  $\mathcal{G} = 0$ ) and **gain\_pos\_of\_nonconstant** (a non-constant reward gives  $\mathcal{G} > 0$ ). We claim nothing beyond these two facts; they suffice to organise the evidence. Content and intent, we will see, present the selector with high reward spread and yield selection gains; the paralinguistic read-out of this frozen encoder presents little spread and yields none.

## 5.2 Content/Intent Selection: High Spread, Real Gains

Our positive result is a frozen-bi-encoder *selection* gain on spoken-language-understanding intent. The mechanism is fixed and simple: the frozen **omni-embed-nemotron-3b** encoder embeds the audio query and a set of candidate *cards* (textual schemas describing each tool/intent), and we select the card of maximum cosine similarity under the verifiable intent-match reward. Nothing is decoded and nothing is trained—the “policy” is the choice of candidate-card representation the selector ranks. We stress this attribution because it is the whole of the claim: these are *frozen bi-encoder verifiable-selection* gains over candidate cards, *not* generative candidates, best-of- $N$  decoding, or any generative operator, none of which we ran.

On **MInDS-14** (en-US tool/intent,  $n=182$ , seed 42), Table 1 reports three candidate-card representations under the identical selection rule. A naive card reaches  $\text{Acc}@1 = 0.720$ ; a raw-schema card raises this to 0.857; and the policy card—a boundary-aware schema (name, description, examples, boundary notes)—reaches 0.984. The policy improvement over the raw-schema baseline is +0.126, with a 95% paired-bootstrap CI of [0.077, 0.181] ( $n=182$ , seed 42). The gain is well separated from zero: the selector is exploiting real reward spread across the candidate cards. **SLURP intent** corroborates the same mechanism on a second corpus: the identical frozen-bi-encoder card-selection move improves intent accuracy from 0.550 to 0.880 (+0.330). Both numbers are single-frozen-model selection gains and are attributed as such.

Reproduce with the committed command against the committed artifact `_repro/minds14_toolintent_paired.json`. We add an honest contamination caveat: SLURP and MInDS-14 are public benchmarks that may overlap pretraining corpora, so the absolute accuracies should be read with that risk in mind; the *contrast* between candidate-card representations under a fixed selector is the load-bearing quantity, and it is a within-model comparison.

We differentiate this result honestly from adjacent best-of- $N$ /reranking work in this series, which targets ASR. The novelty here is narrow and specific: a *frozen omni-embedding* encoder used for SLU/intent *selection* over candidate cards, not ASR best-of- $N$  decoding. The value is the demonstration that a high-spread verifiable reward produces a real, reproducible selection gain on a single frozen speech encoder.

Table 2: Frozen omni content read-out on CREMA-D (linear/kNN probe, dev-selected layer, 3 seeds; committed artifact `_repro/paralinguistic_negative_probe.json`). The content embedding carries near-chance speaker and only modest emotion information: little paralinguistic spread for a selector to exploit.

| Factor  | Ways | Chance | Mean acc | $\times$ chance |
|---------|------|--------|----------|-----------------|
| Speaker | 91   | 0.011  | 0.033    | $3.0\times$     |
| Emotion | 6    | 0.167  | 0.405    | $2.4\times$     |

Table 3: CREMA-D emotion, training-free pool-selection (mean-pool baseline vs. attentive-statistics pooling, both layers dev-selected, paired-delta bootstrap on TEST, five seeds; committed artifact `_repro/emotion_pool_paired_v2.json`). The across-seed 95%  $t$ -CI  $[-0.043, +0.116]$  spans zero: no significant emotion gain. The earlier single-seed  $+0.097$  was the best seed under oracle test-layer selection, reproduced here only as the best of five.

| Seed  | base (dev-sel) | sel (dev-sel)  | $acc_{base}$ | $acc_{sel}$ | $\Delta$ | paired 95% CI                                           |
|-------|----------------|----------------|--------------|-------------|----------|---------------------------------------------------------|
| 42    | audio-mean L0  | audio_attn L16 | 0.400        | 0.497       | $+0.097$ | $[+0.027, +0.167]$                                      |
| 7     | audio-mean L0  | audio_attn L0  | 0.423        | 0.370       | $-0.053$ | $[-0.117, +0.010]$                                      |
| 123   | audio-mean L0  | audio_attn L16 | 0.390        | 0.473       | $+0.083$ | $[+0.013, +0.150]$                                      |
| 2024  | audio-mean L0  | audio_attn L16 | 0.493        | 0.487       | $-0.007$ | $[-0.077, +0.063]$                                      |
| 31337 | audio-mean L0  | audio_attn L16 | 0.400        | 0.463       | $+0.063$ | $[-0.010, +0.130]$                                      |
| mean  | —              | —              | —            | —           | $+0.037$ | across-seed 95% $t$ -CI $[-0.043, +0.116]$ (spans 0; nu |

### 5.3 Paralinguistics: Low Spread, No Gain

The negative result is equally concrete and, we argue, equally a contribution: a precise committed measurement plus the spread explanation for it. Two facts.

First, the frozen omni *content* read-out carries little paralinguistic information. On CREMA-D, with the read-out layer selected on dev and three seeds, speaker identity is recoverable at *near chance*—mean accuracy 0.033 on a 91-way task ( $3.0\times$  chance, chance 0.011)—and emotion only modestly—mean accuracy 0.405 on a 6-way task ( $2.4\times$  chance, chance 0.167) (committed artifact `_repro/paralinguistic_negative_probe.json`). Table 2 summarises. For a *content* embedding, near-chance speaker recovery is close to definitional rather than surprising; the point is that an index built on this frozen omni content embedding would retrieve on content, not on who-spoke or how-they-felt.

Second, training-free selection over this encoder yields *no significant* paralinguistic gain. Treating the choice of read-out (layer and pooling) as the selectable degree of freedom, and evaluating the emotion improvement of attentive-statistics pooling over a mean-pool baseline—both layers selected on dev and frozen before test, with a paired-delta bootstrap and five seeds  $\{42, 7, 123, 2024, 31337\}$ —the across-seed mean improvement is  $\Delta = +0.037$  with a 95%  $t$ -CI of  $[-0.043, +0.116]$ , which *spans zero* (committed artifact `_repro/emotion_pool_paired_v2.json`). Table 3 reports the per-seed detail. The headline is a null: training-free pool-selection produces no significant emotion gain. We note plainly that an earlier single-seed report of  $+0.097$  was the best seed under *oracle test-layer* selection (a selection leak); under dev-selection it reproduces only as the best of five seeds, and the appropriate across-seed aggregate is the null above.

We keep the scope of this negative tight. It is a linear/kNN probe and a pool-selection experiment on *one* frozen content bi-encoder; it does not generalise to frozen omni models as a class, and CREMA-D content is a 12-sentence closed read set, so we make no content claim from it. What it establishes is precisely what the spread lens predicts: this content encoder presents the selector with little paralinguistic spread, so training-free selection over it has little to exploit.

## 5.4 Reading the Evidence

The two results are one story. The gain of training-free selection tracks the reward *spread* across candidates, exactly as the OSA-1 identity  $\mathcal{G} = \beta \text{KL}(\|q_0\| \|q^*\)$  and its two Lean endpoints (`flat_no_gain`, `gain_pos_of_nonconstant`) describe. Content and intent give the selector high spread—the candidate cards differ substantially in verifiable reward—and we obtain real, reproducible single-model gains (+0.126 [0.077, 0.181] on MInDS-14; +0.330 corroborating on SLURP). The paralinguistic read-out of the same frozen encoder gives the selector little spread, and we obtain no significant gain (emotion across-seed CI [−0.043, +0.116]; speaker near chance). We report both intervals in full and state the negative as plainly as the positive.

A brief note on what an *agent-level* extension would require, to be honest about scope rather than to claim it. Moving beyond single-model selection to a genuinely compositional, cross-session program would need (a) a cross-session same-speaker corpus, which our frozen data does not contain; (b) a locally-driven generative operator, which we never ran; and (c) a genuinely new non-separable irreducibility result. Absent these, the spread lens is already informative: on a fixed reward, mere context isolation adds no spread and therefore no gain. We leave that program to future work and rest this paper on the two measured results above.

## 6 Threats to Validity and Limitations

We hold this work to the standard of self-criticism it advocates, and enumerate the threats in descending order of how much they could change the paper’s claims. The scope is deliberately narrow: a *single frozen speech model*, with no weight update and no structural change, and “training-free RL” realised concretely as inference-time *verifiable-reward selection* — a frozen bi-encoder (omni-embed-nemotron-3b, a SentenceTransformer) ranking candidate cards by cosine under a verifiable reward. Every reproducible result below is of that one mechanism; we never ran a local generative operator, and we make no claim about one. Our two load-bearing empirical findings are a *positive* selection gain where the reward has spread across candidates (MInDS-14 tool/intent, Section 5) and a *negative* one where it does not (the paralinguistic read-out).

**Possible pretraining contamination of the public content/intent sets.** Our positive results are on public spoken-language-understanding corpora (MInDS-14 en-US, SLURP intent) whose transcripts and label schemata may have entered the pretraining data of the frozen encoder. A high Acc@1 on such sets can in part reflect memorised surface form rather than an activated latent skill, and inference-time selection against a verifiable reward is, by construction, an over-optimization engine that will happily exploit any such regularity [gao, 2022, ska, 2022, spu, 2025]. We mitigate by reporting a *paired* contrast against a raw-schema baseline on the same items and by committing the artifact (`_repro/minds14_toolintent_paired.json`: Acc@1 0.984 vs. 0.857,  $\Delta = +0.126$ , 95% paired-bootstrap CI [0.077, 0.181],  $n = 182$ , seed 42), so the gain is measured as a within-item improvement rather than an absolute score. But a contamination-controlled split [con, 2024, gor] would be needed to fully exclude leakage, and until then the content/intent gains should be read as gains *relative to the baseline read-out*, not as evidence of uncontaminated generalisation.

**The negative result is specific to *this* frozen content encoder, not to frozen models as a class.** Our negative is precise and committed, but narrowly scoped. On CREMA-D, a linear/kNN probe of the frozen omni-embed *content* read-out (dev-selected layer, three seeds, `_repro/paralinguistic_negative_probe.json`) sits near chance for speaker (mean accuracy 0.033,  $3.0\times$  the 91-way chance rate) and is only modest for emotion (0.405,  $2.4\times$  the 6-way chance rate); and training-free pool-selection over that read-out yields *no* significant emotion gain (across-seed 95% CI [−0.043, +0.116], spanning zero, `_repro/emotion_pool_paired_v2.json`). This bounds *that read-out* and nothing more. A purpose-built speaker encoder (e.g. an ECAPA-TDNN embedding) would carry speaker identity well; that is a well-established property of dedicated speaker models, not something we tested here, and our claim is only that the *omni content read-out* affords little paralinguistic spread to select over. The finding is thus a statement about where verifiable-reward selection over this particular frozen encoder has purchase, consistent with evidence that omni models tend to “read, not listen” on paralinguistic cues [vox, 2026, lis, 2025] and that speaker identity resists in-context correction [spk, 2026].

**Selection is not generation; we did not run a local generative operator.** Everything we report is *selection*: a frozen bi-encoder reranking a fixed pool of candidate cards. We never instantiated a locally-driven generative policy that could *produce* candidates beyond the pool, so we cannot speak to whether generation would recover paralinguistic signal that selection cannot. The correct attribution of every number here is “frozen bi-encoder verifiable selection over candidate cards,” not best-of- $N$  decoding from a generative model; conflating the two would overstate what we demonstrated.

**Single-corpus, limited-seed intervals.** The positive result rests on one corpus per task with modest evaluation sets, where bootstrap intervals [bis] are wide and statistical power is low [car, 2020, dro, hen, 2017]. The SLURP corroboration (0.550  $\rightarrow$  0.880, +0.330, same frozen-bi-encoder card-selection mechanism) is documented but on a single split. The negative rests on three (probe) and five (pooling) seeds on CREMA-D alone. Effects should be read as directional and replicated only within the committed artifacts; any residual mismatch between an in-text number and its artifact should be read as the artifact governing.

**Goodhart pressure, sharpened for a model-dependent reward.** Pushing the pool size  $N$  moves probability mass toward whatever the reward measures, including its defects [gao, 2022, ska, 2022, rlv, 2025]. Verifiable rewards (intent accuracy, exact match) are more robust than learned reward models but are not immune. The sharper risk is that our selection reward is a *cosine computed from the same frozen encoder* that produces the candidate embeddings: it can be high-spread for reasons internal to the model rather than because it tracks the target skill, so a certified gain may measure the model’s self-consistency. Because we change no weights the failure is recoverable (lower  $N$ , decouple the verifier), but the reward should be grounded on held-out, contamination-controlled splits with an externally verifiable target, or the gain is an artifact of the read-out.

**Relation to best-of- $N$  / reranking (honest differentiation).** The individual ingredients — best-of- $N$  and reranking [bei, 2024, ver, 2025], controlled decoding [mud, 2023], and the Gibbs-optimal tilting  $q^* \propto q_0 \exp(R/\beta)$  from alignment-as-inference [kor, 2022, yan, 2024] — are all public, and our positive mechanism is close to that territory. We do not claim the mechanism as novel. What is specific here is the coupling of a *frozen omni-embed encoder* to spoken-language *intent/SLU selection*, together with a precise committed *paralinguistic negative* and the reward-spread lens that explains the contrast — not ASR best-of- $N$ , which is well-trodden elsewhere. The contribution is integrative and empirical, and should be weighted as such.

**Future work: the agent-level program is deferred, not disproved.** A larger cross-session paralinguistic-memory program — one frozen model routing and rewarding another across sessions — does not follow from what we ran, and we defer it honestly. It would require three things we do not have: (a) a cross-session, same-speaker corpus (absent from our frozen data), on which paralinguistic memory could even be measured; (b) a locally-driven generative operator, so that candidates are produced rather than merely selected; and (c) a genuinely new non-separable irreducibility theorem showing that decomposing a task into isolated contexts can add optimization headroom a single model lacks. Absent (c), the spread lens is explicit that context isolation adds *nothing* on a fixed reward: with no new non-constant reward component there is no new spread, and by the identity below no new gain. Establishing that such a component exists, and that it carries spread, is the precondition for any agent-level claim — and it is exactly what our present data do not supply.

## 7 Conclusion

We set out to test one modest, falsifiable claim on a single frozen speech model, changing no weights and no structure: that the gain available to training-free RL — realised here as inference-time *verifiable-reward selection*, a frozen bi-encoder ranking candidate cards by cosine — is governed by the *reward spread* across candidates, and not by any agentic machinery. On the content/intent side the spread is real and large, and selection turns it into gain: on MInDS-14 en-US tool/intent the frozen bi-encoder reaches Acc@1 0.984 against a raw-schema baseline of 0.857, a paired improvement of +0.126 (95% CI [0.077, 0.181],  $n = 182$ ,



seed 42; `_repro/minds14_toolintent_paired.json`), with SLURP intent corroborating ( $0.550 \rightarrow 0.880$ ,  $+0.330$ ) through the same mechanism. On the paralinguistic side the spread is essentially absent on this read-out: the frozen content encoder probes near chance for speaker (mean 0.033,  $3.0\times$  chance) and only modestly for emotion (0.405,  $2.4\times$  chance) on CREMA-D, and pool-selection yields no significant emotion gain (across-seed CI  $[-0.043, +0.116]$ ). We report the negative not as a setback but as a result of equal standing: a precise, committed measurement that scopes exactly where the method works.

The lens that ties these together is deliberately lightweight and used qualitatively only. The identity  $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$  — the free-energy / Donsker–Varadhan characterisation of the gap between what the frozen model already does ( $q_0$ ) and what the reward would have it do ( $q^*$ ) — is a known result, not a novel theorem, and we invoke it to *explain*, not to prove: the gain is bounded by the reward spread over  $\text{supp}(q_0)$ , so a flat reward yields vanishing gain while a high-spread reward yields large gain, on the very same frozen model. Two strict, sorry-free Lean lemmas anchor the qualitative core: a flat reward forces zero gain (`flat_no_gain`), and a non-constant reward forces strictly positive gain (`gain_pos_of_nonconstant`). That is precisely the contrast we observe — content/intent has spread and selection helps; paralinguistics, on this read-out, has little spread and it does not.

The conclusion is therefore modest and, we believe, true. Training-free verifiable-reward selection on one frozen speech model helps for high-spread content and intent, does not help for low-spread paralinguistics on the content read-out we measured, and the reward-spread lens explains why. What we offer is a small, reproducible pair of results and a lens that predicts their sign — not a deployable agentic stack, and not a claim beyond the single model and the committed artifacts we can currently stand behind.

## A Lean 4 Formalization

The lens of this paper — that the optimization gain of training-free RL (inference-time verifiable-reward selection) is governed by the *reward spread*, not by any model class — rests on one qualitative identity, and *that* identity is machine-checked in **Lean 4 + mathlib** (toolchain and mathlib both pinned to v4.31.0; the exact mathlib commit is recorded in the `lake-manifest.json` committed alongside the proof project, see Section A), in the project `proofs/tfrl` (Lean namespace `TfrlProofs`). We are careful about what “machine-checked” means, and Section A below states it precisely. The retained core is small: the master identity and T1 optimality, the flat-reward collapse, and the gain identity  $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$  together with its nonnegativity *and its strict positivity under any non-constant reward* (`gain_eq`, `gain_nonneg`, `gain_pos_of_nonconstant`, `kl_pos_of_ne`, `flat_no_gain`) — all **fully sorry-free**. This is exactly the formal content of the spread lens: the gain is a function of the *reward structure* (the range  $R_{\max} - R_{\min}$  over  $\text{supp}(q_0)$ ), and nothing in the hypotheses references a model class. Two sorry-free facts make the lens precise. (i) `gain_pos_of_nonconstant`: every non-degenerate verifiable reward contributes *strictly* positive gain, even on a single frozen model — this is why the high-spread content/intent rewards yield selection gains. (ii) `flat_no_gain`: a flat reward yields *zero* gain — this is why a factor with little reward spread (our paralinguistic negative) has nothing for training-free selection to exploit. The Lean project `proofs/tfrl` may contain additional lemmas beyond those listed here; only the lemmas in Section A are load-bearing for this paper.

We are explicit about the boundary between the machine-checked identity and its quantitative companion. The qualitative identity  $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$  and the strict lemmas above are machine-checked **sorry-free**. The *quantitative* spread bound  $\mathcal{G} \leq \text{spread}^2 / (8\beta)$  is the standard Donsker–Varadhan / Hoeffding statement (proved on paper in Lemma 1); we use it *qualitatively* — to say “small spread  $\Rightarrow$  small headroom” — and we do **not** bill it as a novel or machine-checked ceiling. Throughout,  $\mathcal{Z}$  is a **Fintype** (the inference-time action space),  $q_0 : \mathcal{Z} \rightarrow \mathbb{R}$  is a strictly positive reference distribution with  $\sum_z q_0(z) = 1$ ,  $R : \mathcal{Z} \rightarrow \mathbb{R}$  is a reward, and  $\beta > 0$  a temperature.

**Verification status.** To prevent any over-reading of “machine-checked”, we tabulate the exact status of each retained Lean result.

| Lean lemma                                     | Status     | Undischarged input |
|------------------------------------------------|------------|--------------------|
| <code>kl_nonneg</code>                         | sorry-free | —                  |
| <code>kl_pos_of_ne</code> (strict Gibbs)       | sorry-free | —                  |
| <code>F_sub_eq_beta_mul_kl</code>              | sorry-free | —                  |
| <code>tilting_optimal</code> (T1)              | sorry-free | —                  |
| <code>qstar_eq_q0_of_const</code> (T3)         | sorry-free | —                  |
| <code>gain_eq</code> (OSA-1a)                  | sorry-free | —                  |
| <code>gain_nonneg</code> (OSA-1b)              | sorry-free | —                  |
| <code>gain_pos_of_nonconstant</code> (OSA-1b') | sorry-free | —                  |
| <code>flat_no_gain</code> (OSA-1c)             | sorry-free | —                  |

Every entry compiles with no axioms beyond mathlib’s and no `sorry`; the exact mathlib commit against which they compile is pinned in the `lake-manifest.json` committed with the `proofs/tfml` project, so the build is reproducible bit-for-bit. (“OSA” abbreviates *Optimization-Space Adequacy*; only the OSA-1 single-block lemmas are used in this paper.) The one quantitative statement we invoke — the  $\text{spread}^2/(8\beta)$  ceiling — is **not** a Lean result: it is the standard Hoeffding bound, proved on paper in Lemma 1 and used only qualitatively.

**Tilting primitives and the Gibbs inequality.** The exponentially tilted optimum  $q^*$ , its partition function  $Z$ , and the KL-regularized objective  $F(\cdot)$  are defined exactly as in the main text. We reproduce the Lean signatures verbatim.

```
noncomputable def Zpart (q0 R : Z → ℝ) (β : ℝ) : ℝ := ∑ w, q0 w * Real.exp (R w / β)
noncomputable def qstar (q0 R : Z → ℝ) (β : ℝ) (z : Z) : ℝ :=
  q0 z * Real.exp (R z / β) / Zpart q0 R β
noncomputable def F (q0 R : Z → ℝ) (β : ℝ) (q : Z → ℝ) : ℝ :=
  (∑ z, q z * R z) - β * ∑ z, q z * Real.log (q z / q0 z)
```

```
theorem kl_nonneg {p r : Z → ℝ} (hp : ∀ z, 0 ≤ p z) (hr : ∀ z, 0 < r z)
  (hpsum : ∑ z, p z = 1) (hrsum : ∑ z, r z = 1) :
  0 ≤ ∑ z, p z * Real.log (p z / r z)
```

`kl_nonneg` is Gibbs’ inequality: the pointwise estimate  $p(z) - r(z) \leq p(z) \log(p(z)/r(z))$  (trivial when  $p(z) = 0$ ; otherwise  $\log x \leq x - 1$  with  $x = r(z)/p(z)$  via `Real.log_le_sub_one_of_pos`) is summed only over  $\text{supp}(p)$ . On that index set  $\sum_{z \in \text{supp}(p)} p(z) = 1$ , while  $\sum_{z \in \text{supp}(p)} r(z) \leq \sum_z r(z) = 1$  (the partial sum over the support is  $\leq$  the full sum, since  $r \geq 0$ ); this gives  $\sum_{z \in \text{supp}(p)} (p(z) - r(z)) \geq 1 - 1 = 0$ , hence  $\text{KL}(p \| r) \geq 0$ . The one-sided bound “ $\leq$  the full sum” is sufficient.

We additionally prove the *strict* form, `sorry-free`:

```
theorem kl_pos_of_ne {p r : Z → ℝ} (hp : ∀ z, 0 < p z) (hr : ∀ z, 0 < r z)
  (hpsum : ∑ z, p z = 1) (hrsum : ∑ z, r z = 1) (hne : ∃ z, p z ≠ r z) :
  0 < ∑ z, p z * Real.log (p z / r z)
```

`kl_pos_of_ne` upgrades Gibbs to its equality case:  $\text{KL}(p \| r) = 0 \iff p = r$ , so  $p \neq r \Rightarrow \text{KL}(p \| r) > 0$ . The pointwise slack  $p(z) \log(p(z)/r(z)) - (p(z) - r(z)) \geq 0$  is *strictly* positive at any  $z$  with  $p(z) > 0$  and  $p(z) \neq r(z)$  (strict form of  $\log x < x - 1$  for  $x \neq 1$ , via `Real.log_lt_sub_one_of_ne`); the hypothesis  $\exists z, p(z) \neq r(z)$  (with  $p$  strictly positive everywhere) supplies such a  $z$ , so summing the nonnegative slacks — at least one strictly positive — gives a strictly positive total. This strict inequality is the engine of the strict gain lemma below: it is exactly why a non-constant reward yields *strictly* positive gain rather than merely nonnegative gain.

**Master identity and T1.** The decomposition  $F(q^*) - F(q) = \beta \text{KL}(q \| q^*)$  is the engine of optimality.

```

theorem F_sub_eq_beta_mul_kl [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  {q : Z → ℝ} (hq : ∀ z, 0 ≤ q z) (hqsum : ∑ z, q z = 1) :
  F q0 R β (qstar q0 R β) - F q0 R β q = β * ∑ z, q z * Real.log (q z / qstar q0 R β z)

```

```

theorem tilting_optimal [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  {q : Z → ℝ} (hq : ∀ z, 0 ≤ q z) (hqsum : ∑ z, q z = 1) :
  F q0 R β q ≤ F q0 R β (qstar q0 R β)

```

**T1** (`tilting_optimal`) is then immediate:  $F(q^*) - F(q) = \beta \text{KL}(q \| q^*) \geq 0$  by `kl_nonneg` and  $\beta > 0$ .

**T3 — flat reward, no tilt.**

```

theorem qstar_eq_q0_of_const [Nonempty Z] {q0 : Z → ℝ} {β c : ℝ}
  (hq0sum : ∑ z, q0 z = 1) {R : Z → ℝ} (hR : ∀ z, R z = c) :
  ∀ z, qstar q0 R β z = q0 z

```

When  $R(z) \equiv c$ ,  $Z = \exp(c/\beta)$  collapses and  $q^*(z) = q_0(z) \exp(c/\beta) / \exp(c/\beta) = q_0(z)$ : a degenerate action space with zero reward spread admits no steering.

**The gain identity** (`TfmlProofs.OptSpace`). We define the optimization gain  $\mathcal{G} := F(q^*) - F(q_0)$  and prove its structural properties. Every lemma below compiles `sorry-free`.

```

noncomputable def gain (q0 R : Z → ℝ) (β : ℝ) : ℝ :=
  F q0 R β (qstar q0 R β) - F q0 R β q0

```

```

theorem gain_eq [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  (hq0sum : ∑ z, q0 z = 1) :
  gain q0 R β = β * Real.log (Zpart q0 R β) - ∑ z, q0 z * R z

```

```

theorem gain_nonneg [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  (hq0sum : ∑ z, q0 z = 1) : 0 ≤ gain q0 R β

```

```

theorem gain_pos_of_nonconstant [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  (hq0sum : ∑ z, q0 z = 1) (hR : ∃ z w, R z ≠ R w) :
  0 < gain q0 R β

```

```

theorem flat_no_gain [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  (hq0sum : ∑ z, q0 z = 1) {c : ℝ} (hR : ∀ z, R z = c) : gain q0 R β = 0

```

`gain_eq` rewrites the gain as  $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R]$  (**OSA-1a**); equivalently, read at  $q = q_0$ , the master identity gives the free-energy / Donsker–Varadhan form  $\mathcal{G} = \beta \text{KL}(q_0 \| q^*)$ . `gain_nonneg` (**OSA-1b**) is Jensen / Gibbs. `gain_pos_of_nonconstant` (**OSA-1b'**) is the strict companion: if the reward takes two distinct values,  $R(z_1) \neq R(z_2)$ , then  $q^* \neq q_0$  (the tilt is non-trivial), so by the strict Gibbs lemma `kl_pos_of_ne` (applied with the strictly-positive argument  $r = q^* > 0$ ) the identity  $\mathcal{G} = \beta \text{KL}(q_0 \| q^*)$  is *strictly* positive. This is the formal statement of the spread lens at the single-block level: **any** non-degenerate verifiable reward yields strictly positive gain — crucially, *even on a single frozen model*, since nothing in the hypotheses references a model class. Conversely `flat_no_gain` (**OSA-1c**) recovers T3: a flat reward (e.g. a frozen embedding read-out that does not separate the labelled classes, hence a degenerate  $\mathcal{Z}$  under that reward) affords *zero* improvement. Together OSA-1b'/OSA-1c make the dichotomy sharp and model-agnostic: gain is governed by whether the reward is non-constant. This is precisely the contrast the paper reports empirically — content/intent rewards have large spread (selection gains), the paralinguistic reward on this frozen encoder is near-flat (no significant gain).

## B Extended Proof: the spread bound

We record, on paper, the standard analytic statement behind the spread lens: the reward spread caps the gain. This is the classical Donsker–Varadhan / Hoeffding bound; we present it for completeness and use it only qualitatively. It is *not* a Lean result and we do not bill it as one.

## B.1 Hoeffding instance: the gain ceiling $\text{spread}^2/(8\beta)$

**Lemma 1** (Bounded-reward gain ceiling). *Let  $R : \mathcal{Z} \rightarrow [R_{\min}, R_{\max}]$  with reward spread  $\text{spread} := R_{\max} - R_{\min}$ , and let  $q_0$  be any distribution on  $\mathcal{Z}$  with  $\mathbb{E}_{q_0}$  the corresponding expectation. Then for every  $\beta > 0$ ,*

$$\mathcal{G}(q_0, R, \beta) = \beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R] \leq \frac{\text{spread}^2}{8\beta}.$$

*Proof.* Let  $X := R(z)/\beta$  for  $z \sim q_0$ , supported in  $[a, b]$  with  $a = R_{\min}/\beta$ ,  $b = R_{\max}/\beta$ , so  $b - a = \text{spread}/\beta$ . Set  $\varphi(\lambda) := \log \mathbb{E}[e^{\lambda X}]$ ; then  $\varphi(0) = 0$ ,  $\varphi'(0) = \mathbb{E}X$ , and  $\varphi''(\lambda) = \text{Var}_\lambda(X)$  under the tilted law  $d\mu_\lambda \propto e^{\lambda x} d\mu$ . Since  $X \in [a, b]$  almost surely, the same holds  $\mu_\lambda$ -a.s., and a random variable bounded in an interval of length  $b - a$  has variance at most  $(b - a)^2/4$  (extremal case: the two-point distribution at the endpoints). Hence  $\varphi''(\lambda) \leq (b - a)^2/4$ . By Taylor’s theorem with the Lagrange remainder, for some  $\xi$  between 0 and  $\lambda$ ,

$$\varphi(\lambda) = \varphi(0) + \lambda\varphi'(0) + \frac{1}{2}\lambda^2\varphi''(\xi) \leq \lambda\mathbb{E}X + \frac{\lambda^2(b - a)^2}{8},$$

so  $\psi(\lambda) := \varphi(\lambda) - \lambda\mathbb{E}X \leq \lambda^2(b - a)^2/8$ , which is Hoeffding’s lemma. Taking  $\lambda = 1$ ,

$$\log \mathbb{E}[e^X] - \mathbb{E}[X] = \psi(1) \leq \frac{(b - a)^2}{8} = \frac{\text{spread}^2}{8\beta^2}.$$

Multiplying by  $\beta > 0$  and substituting  $X = R/\beta$  gives  $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R] \leq \text{spread}^2/(8\beta)$ . By `gain_eq` the left side is exactly  $\mathcal{G}$ .  $\square$

Lemma 1 makes the scaling explicit: the inference-time selection gain is quadratically throttled by the reward spread,  $\mathcal{G} = O(\text{spread}^2/\beta)$ , so a flat or near-flat reward landscape (T3, **OSA-1c**) yields vanishing headroom regardless of candidate budget — the analytic counterpart of reward over-optimization saturation [gao, 2022]. Together with `gain_pos_of_nonconstant` (strict lower bound for any non-constant reward), it brackets the gain from both sides: it is the reward spread, not the model class, that sets the ceiling. We use this bound qualitatively only.

## C Notation

| Symbol                     | Meaning                                                                         |
|----------------------------|---------------------------------------------------------------------------------|
| $\mathcal{Z}$              | inference-time action space (candidate cards; no weights change)                |
| $q_0$                      | base reference policy $q_0$ (strictly positive, $\sum_z q_0(z) = 1$ )           |
| $q$                        | a search distribution $q$ over $\mathcal{Z}$                                    |
| $q^*$                      | Gibbs / tilted optimum $q^*(z) = q_0(z)e^{R(z)/\beta}/Z$                        |
| $R$                        | verifiable reward function $\mathcal{Z} \rightarrow \mathbb{R}$                 |
| $\beta$                    | temperature ( $> 0$ ); KL-regularization strength                               |
| $Z$                        | partition function $\sum_w q_0(w)e^{R(w)/\beta}$                                |
| $F(q)$                     | KL-regularized objective $\mathbb{E}_q[R] - \beta \text{KL}(q \parallel q_0)$   |
| $\text{KL}(p \parallel q)$ | Kullback–Leibler divergence                                                     |
| $\mathbb{E}$               | expectation                                                                     |
| $\mathcal{G}$              | optimization gain $F(q^*) - F(q_0) = \beta \text{KL}(q_0 \parallel q^*) \geq 0$ |
| spread                     | reward spread $R_{\max} - R_{\min}$                                             |
| $N$                        | number of candidate cards in verifiable selection (best-of- $N$ / reranking)    |

## D Verifiable Reward Functions

The reward callables used as verifiable signals  $R$  are exposed by the shared library `speechrl_common.rl`, and are deliberately deterministic and contamination-resistant [tul, 2024]. Heavy dependencies (`jiwer`, `scikit-learn`) are imported lazily inside each function so the package and its smoke tests import without the full stack. In our experiments all selection is performed by a *frozen bi-encoder* ranking candidate cards by cosine under one of these rewards; no weights are updated.

**Content / ASR (`rl.reward`).** `wer(reference, hypothesis)` returns the word error rate after normalization; `asr_reward` returns  $\max(0, 1 - \text{WER}) \in [0, 1]$ , a bounded reward whose spread directly governs the gain ceiling of Lemma 1; `exact_match_reward` returns  $\mathbf{1}[\text{normalized strings equal}] \in \{0, 1\}$  for closed-form tasks (intent, tool selection, keyword). These content/intent rewards are the ones for which our experiments realize a large, non-degenerate spread over the candidate cards — consistent with the frozen bi-encoder card-selection gains reported in the main text (MInDS-14 tool/intent, SLURP intent).

**Retrieval / embedding (`rl.embedding_metrics`).** `recall_at_k(query, gallery, labels_q, labels_g, k)` and `mean_reciprocal_rank(...)` score the frozen model’s pooled embeddings against a labelled gallery under cosine similarity (`cosine_sim_matrix` with  $\ell_2$  normalization); `retrieval_reward(..., k)` packages `recall@k` as a  $[0, 1]$  reward. We flag honestly that for the *speaker* factor this reward is near-degenerate on the frozen omni embedding: the content read-out carries near-chance speaker identity (mean accuracy 0.033,  $3.0\times$  chance on a 91-way task; dev-selected layer, 3 seeds; committed artifact `_repro/paralinguistic_negative_probe.json`). This is a near-flat landscape under which Lemma 1 predicts vanishing gain — there is little paralinguistic spread for training-free selection to exploit.

**Probe accuracy (`rl.probe`).** `linear_probe_accuracy / knn_probe_accuracy` fit a frozen-feature classifier (held-out split) to score readability of the emotion (SER) and language/intent factors from the embeddings; `probe_reward(..., kind)` returns the held-out accuracy as the reward. For the *emotion* factor the frozen read-out is only modestly above chance (accuracy 0.405,  $2.4\times$  chance on a 6-way CREMA-D task; dev-selected layer, 3 seeds; same committed artifact `_repro/paralinguistic_negative_probe.json`), and training-free pool-selection over this encoder yields *no significant gain*: a multi-seed re-run with dev-selected pooling layer (5 seeds: 42, 7, 123, 2024, 31337) gives mean  $\Delta = +0.037$  with a 95% across-seed *t*-CI of  $[-0.043, +0.116]$ , which spans 0 — not significant (committed artifact `_repro/emotion_pool_paired_v2.json`).<sup>1</sup> The previously-reported single-seed  $+0.097$  was the best seed under oracle test-layer selection (a selection-leak / winner’s-curse artifact). We scope this negative precisely: it is established on *one* frozen content bi-encoder (omni-embed) via a linear/kNN probe and pool-selection over its embedding space, and it says only that training-free selection over *this* encoder has little paralinguistic spread to exploit. Auxiliary metrics (`macro_f1`, `bleu`, `chrf`, `eer`) in `rl.metrics` are used for evaluation only, not as RL rewards, to avoid coupling the reward and the test statistic [ska, 2022].

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<sup>1</sup>The per-seed paired-delta bootstrap excludes 0 in 2/5 seeds; this vote-count is reported only for completeness and is not the headline statistic.



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